🌿 AI Plant Disease Detection

An AI-powered plant disease detection system that runs entirely in **Google Colab**. Upload a photo of a plant leaf and get an instant diagnosis with treatment recommendations.

Show Image

Show Image

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🚀 Quick Start

1. Click the badge below to open in Colab:

   Show Image

2. Run all cells top to bottom (`Runtime → Run all`)
3. Upload your plant leaf image when prompted
4. Get your diagnosis instantly!

🔬 What It Does

Step	Description
📷 Upload	User uploads one or more leaf images directly in Colab
🧠 Inference	MobileNetV2 backbone classifies the image into 38 disease classes
📊 Visualise	Top-3 predictions shown as a confidence bar chart
💊 Treatment	Disease name, recommended treatment, medicine, and care advice displayed

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🌱 Supported Plants & Diseases (38 Classes)

Plant	Conditions Detected
🍎 Apple	Apple Scab, Black Rot, Cedar Apple Rust, Healthy
🌽 Corn	Cercospora Leaf Spot, Common Rust, Northern Leaf Blight, Healthy
🍇 Grape	Black Rot, Esca, Leaf Blight, Healthy
🍅 Tomato	Bacterial Spot, Early Blight, Late Blight, Leaf Mold, Septoria Leaf Spot, Spider Mites, Target Spot, Yellow Leaf Curl Virus, Mosaic Virus, Healthy
🥔 Potato	Early Blight, Late Blight, Healthy
🫑 Pepper	Bacterial Spot, Healthy
🍑 Peach	Bacterial Spot, Healthy
+ more	Blueberry, Cherry, Orange, Raspberry, Soybean, Squash, Strawberry

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🛠️ Tech Stack

- **Model**: MobileNetV2 (ImageNet pretrained backbone)
- **Fine-tuning**: PlantVillage dataset (~54,000 images, 38 classes)
- **Framework**: TensorFlow / Keras
- **Runtime**: Google Colab (free GPU supported)
- **Visualization**: Matplotlib

📁 Project Structure

```
plant-disease-detection/
│
├── plant_disease_detection.ipynb   # Main Colab notebook
└── README.md                       # This file
```

🔌 Using Your Own Model

In **Cell 3** of the notebook, three options are provided:

```python
# Option A: Mount Google Drive and load .h5
from google.colab import drive
drive.mount('/content/drive')
model = tf.keras.models.load_model('/content/drive/MyDrive/your_model.h5')

# Option B: Download via gdown (Google Drive public link)
import gdown
gdown.download('https://drive.google.com/uc?id=YOUR_FILE_ID', 'model.h5')
model = tf.keras.models.load_model('model.h5')

# Option C (default): ImageNet backbone (no fine-tuning)
model = build_model()
```

📊 Model Performance

Metric	Value (fine-tuned on PlantVillage)
Accuracy	~96%
Classes	38
Input Size	224 × 224 px

> ⚠️ Default model uses ImageNet weights only. Connect a fine-tuned checkpoint for accurate predictions.

📜 Dataset

PlantVillage Dataset — open-access dataset of 54,306 healthy and diseased plant leaf images across 14 crop species.

🤝 Contributing

Pull requests are welcome! For major changes, please open an issue first.

📄 License

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